

Master Construction Blueprint: Causal-Informed Recommendation System

Project Title: Causal-Informed Recommendation System for Mitigating Echo Chambers

Goal: Build a recommendation engine that moves from reactive prediction (predicting clicks) to proactive intervention (optimizing for long-term interest diversity).

1. System Architecture: The "Causal-RL Loop"

The system architecture is a closed loop where causal inference informs the reward signal for a reinforcement learning agent.

High-Level Data Flow

1. **Input:** User interaction logs (e.g., MIND Dataset).
2. **Module 1: Causal Discovery ("The Brain"):** Learns a Structural Causal Model (DAG) to understand how user features, item features, and recommendations interact causally, not just correlationally.
3. **Module 2: Counterfactual Engine ("The Simulator"):** Queries the SCM to ask: *"What would happen to the user's diversity if I showed Item X instead of Item Y?"*
4. **Module 3: RL Policy ("The Decision Maker"):** An Actor-Critic agent uses the signal from Module 2 to balance short-term clicks with long-term diversity.
5. **Output:** A ranked list of news articles that breaks filter bubbles.

2. Step-by-Step Build Instructions

Phase 1: Environment & Data Setup

Objective: Prepare the workspace and the MIND dataset for causal analysis.

- **Tech Stack:** Python 3.9+, Pandas, NumPy, Scikit-learn.
- **Dataset:** Microsoft News Dataset (MIND).

Instruction Details:

1. **Data Acquisition:** Load the MIND dataset (users, news, behaviors).
2. **Variable Extraction:** Create a processing pipeline to extract four specific tensor sets:
 - **\$U\$ (User Features):** History embedding, dwell time, click counts.
 - **\$I\$ (Item Features):** Category, sentiment score, entity embeddings.
 - **\$A\$ (Action/Treatment):** The recommended item ID.
 - **\$Y\$ (Outcome):** Binary click label AND a computed Interest_Diversity_Score.
3. **Diversity Metric:** Implement a function to calculate Interest_Diversity_Score (e.g., 1 -

Cosine Similarity between candidate item and user history).

Prompt for LLM/Coder: "Write a Python script using Pandas to load the MIND dataset. Extract user features (history), item features (category, sentiment), and interaction labels. Define a 'Homogeneity Score' function that calculates the cosine similarity between a candidate news item and the user's history vector."

Phase 2: Causal Discovery (The SCM)

Objective: Build a Directed Acyclic Graph (DAG) that maps cause-and-effect.

- **Tech Stack:** causal-learn or dowhy.
- **Algorithm:** Hill-Climbing (HC) with Bayesian Information Criterion (BIC).

Instruction Details:

1. **Define Domain Constraints (Tiers):** Enforce a hierarchy to prevent illogical edges (e.g., a click cannot cause a user's age).
 - *Tier 1:* Static User Features (\$U\$)
 - *Tier 2:* Item Features (\$I\$)
 - *Tier 3:* Action (\$A\$)
 - *Tier 4:* Outcome (\$Y\$)
2. **Structure Learning:** Run the Hill-Climbing algorithm on the observational data to learn the DAG structure.
3. **Validation:** Output the Adjacency Matrix and visualize the graph.

Prompt for LLM/Coder: "Using the causal-learn library, implement a constraint-based Hill-Climbing algorithm. Enforce a tiered structure where User Features are parents of Actions, and Actions are parents of Outcomes. Output the final Adjacency Matrix representing the Causal Graph."

Phase 3: The Counterfactual Engine

Objective: Create a distinct module that estimates the "diversity reward" for the RL agent.

- **Tech Stack:** PyTorch or TensorFlow.

Instruction Details:

1. **Input Interface:** The engine must accept a User Vector $\$u\$$ and a Candidate Item Vector $\$i\$$.
2. **Intervention Logic:** It must answer the counterfactual question: $\$E[Y_{\{diversity\}} | do(A=i'), U=u]\$$.
3. **Estimator:** Implement a Propensity Score Matching or a Deep Learning estimator (like TARNet or Dragonnet) to predict the potential diversity outcome if the intervention were applied.
4. **Output:** A scalar value representing the predicted impact on user diversity.

Prompt for LLM/Coder: "Create a Python class CounterfactualEngine. It should

load the learned Causal Graph. Implement a method `estimate_effect(user_vector, item_vector)` that uses Inverse Probability Weighting to predict the user's future diversity score if that item is recommended."

Phase 4: Reinforcement Learning (The Agent)

Objective: Train the policy that actually picks the recommendations.

- **Tech Stack:** PyTorch (Stable Baselines3 or custom implementation).
- **Architecture:** Actor-Critic (A2C or PPO).

Instruction Details:

1. **MDP Setup:**
 - **State (\$s\$):** User profile vector + current diversity score.
 - **Action (\$a\$):** Selection of a news article from the candidate pool.
2. **Composite Reward Function:** This is the critical component. Implement:
$$r = w \cdot r_{\text{engagement}} + (1-w) \cdot r_{\text{diversity}}$$
 - $r_{\text{engagement}}$: Binary click (1 or 0) or click probability.
 - $r_{\text{diversity}}$: The output from the **Counterfactual Engine** (Phase 3).
 - w : Hyperparameter (e.g., 0.7) balancing the trade-off.
3. **Training:** Train the agent to maximize cumulative r over a session trajectory.

Prompt for LLM/Coder: "Build a PyTorch Actor-Critic RL agent. The `get_reward` function must combine a binary click signal with a continuous diversity signal. The loss function should optimize the policy $\pi(a|s)$ to maximize cumulative reward over a session."

Phase 5: Evaluation & Metrics

Objective: Validate the system offline.

Instruction Details:

1. **Accuracy Metrics:** Calculate NDCG@10 and Precision@10 (standard engagement).
2. **Diversity Metrics:**
 - **Intra-list Diversity (ILD):** Average dissimilarity between recommended items.
 - **Homogeneity Score:** Similarity of recommendations to user history.
3. **Benchmark:** Compare results against a standard Collaborative Filtering baseline. The goal is to show higher Diversity metrics with comparable Accuracy.

3. Developer Checklist

Component	Tool/Algorithm	Input	Output
Data Prep	Pandas/MIND	Raw Logs	Tensors (\$U, I, A,

			Y\$)
Discovery	Hill-Climbing + BIC	Observational Data	Causal DAG Structure
Estimator	Causal Inference	DAG + User + Item	Predicted Diversity Impact
RL Policy	Actor-Critic (PPO/A2C)	User State	Recommended Item List
Reward	Weighted Formula	Click + Diversity Est.	Scalar Reward Signal